

Buttonwood

ART BIBLE

Terraria craft. Minecraft logic. Buttonwood soul.



The approved concept. Native game sprites interpret this direction; the concept is not a sprite sheet.

A place worth building.

A warm, handmade settlement with a readable block world underneath every soft curve.



APPROVED NATIVE ART / Deterministic scene assembled from the existing world generators, at 1 art pixel per PDF point.

01 Warmth before clutter

Cream hats, coral roofs, mint trim and gold buttons make a recognizable family. Give useful shapes room to breathe.

02 The grid is honest

Building and terrain art must respect placement, support and excavation. Ornament can soften the silhouette without rewriting it.

03 Life at small scale

Show work with attached hands, planted feet and purposeful tools. Read the Worker and the action before the decoration.

Borrow the principle. Find our voice.

The references guide decisions. Buttonwood keeps its own palette, proportions, buildings and visual identity.

TERRARIA Craft in a side-view world	MINECRAFT Material and construction logic	BUTTONWOOD A handmade settlement
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OBSERVE

Dense worlds remain identifiable through silhouette, material contrast and depth.	Repeated blocks remain useful construction units with recognizable material identity.	Warm characters and curved roof profiles sit against calm, modular terrain.
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TRANSLATE

Let foreground objects read first; use quiet scenery and a few meaningful highlights.	Preserve the 32-unit cell. Make joins, exposed edges and construction states trustworthy.	Share colored contours, three-tone materials, mint accents and generous quiet areas.
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AVOID

Copying sprites, dense ornament everywhere, or using detail as a substitute for hierarchy.	Importing 3D cube perspective into a front-elevation game, or outlining every terrain cell.	A generic fantasy skin, enlarged pixel scales on bigger buildings, or UI from a different world.
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DESIGN INTERPRETATION / Lessons drawn from official Terraria imagery and Mojang texture resources. Source notes on page 16.

One pixel. One world.

The whole camera may zoom. Individual world assets never get their own scale.

MEASURED NATIVE STUDY



Home 128 x 128

Four blocks high

Forge 96 x 96

Three blocks high

32

world units per block

Adult Worker: 40 x 32 canvas

Ground anchor: (18, 31)

Worker height: approximately 30 px

A Start at 1x

Judge silhouette, materials and eye spacing at native resolution before inspecting larger pixels.

B Draw with integers

Use deliberate connected clusters, stepped curves, one-pixel colored contours and upper-left light.

UI lives on its own canvas. Hand-draw each icon and miniature at 32 x 32. Keep building portraits at their native sprite dimensions.

A small palette, a rich world.

Exact swatches are read from palette.js. Shadow, base and light describe each material with restraint.

Cream

#d8c397 / #f7e8c4

#fff3d6

Mint

#597d70 / #8fb69a

#b9d1ab

Coral

#ad5955 / #d97d68

#f0a17c

Timber

#704939 / #9c684d

#c28d60

Stone

#737e78 / #9b9b91

#c7c3aa

Loam

#5e4236 / #785242

#94694d

Grass

#426f49 / #6d9d58

#a5c975

Water

#397e89 / #5eaaa6

#a5d9c6

Lava

#a95248 / #dd7958

#f5c77f

Deep stone

#4f4d60 / #675f72

#8b8290

Loam

Stone

Timber

Leaves

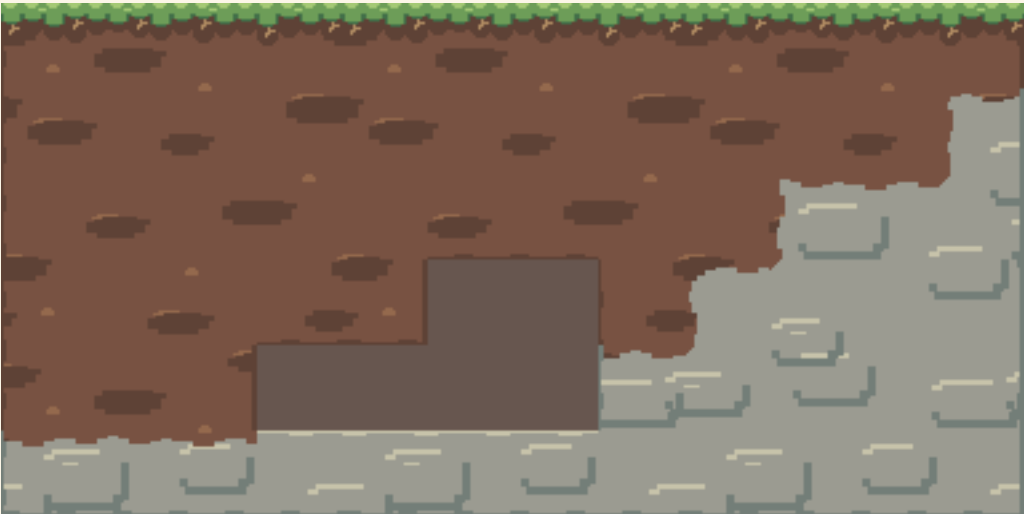
Depth

Obsidian

Quiet material planes.
Use sparse directional marks, shared outlines and deliberate clusters. Add a new ramp only through the shared palette and scene review.

Connected terrain. Honest edges.

Our world is made of cells; its ground should read as continuous soil and stone.



APPROVED NATIVE TERRAIN / 12 x 6 cells. Shared world-coordinate texture, exposed sod, a cave opening and soil-to-stone transitions.

- 1

Same material joins

Hide internal tile borders. Never add a dark outline around each occupied cell.
- 2

Exposed edges

Use short material-colored bevels and controlled corners. Keep the full solid cell silhouette.
- 3

One owner per boundary

Step the transition inside occupied cells. Resolve concave corners as carefully as straight edges.

EDGE REVIEW MATRIX
Solid field / exposed top / outside corner
Concave corner / mixed material / excavation
Placed aligned / moved piece / rotated piece

Visuals follow the simulation. Natural neighbors come from the terrain grid. Only stationary, aligned placed bodies share external joins. Texture never changes support, placement or saves.

Small people. Clear intentions.

Keep the broad cream hat, mint band, short overalls, gold buttons and stable planted feet.



40 x 32 canvas
Ground anchor (18, 31)
One-pixel walk rise at most
Shared proportions; varied skin and hair

APPROVED APPEARANCES / 3x inspection enlargement. These are adults with chibi proportions.

WALK / SIX FRAMES / 110 MS EACH



WORK / SIX GALLERY POSES / 90 MS EACH



ENGINE TRUTH / The renderer starts at impact pose 3 and follows recovery poses 4-5 (270 ms total). Pre-strike anticipation, children and new actions remain deferred.

A family with five distinct jobs.

Keep front elevations, Worker-scale openings and stable foundations. Give each function a memorable opening.



Home

Welcoming doorway



Workshop

Open working bay



Forge

Compact fire mouth

Same art pixel.

Bigger buildings use more canvas area. Their pixels do not get bigger.



Storehouse

Broad loading doors



Blacksmith

Open anvil bay

APPROVED NATIVE SPRITES / Home and Workshop 128 x 128; Forge 96 x 96; Storehouse and Blacksmith 160 x 128.

An icon is a drawing, too.

Preserve the identifying roof, opening and accent. Recompose each building on a new 32 x 32 canvas.



WORLD SPRITE

128 x 128 source, shown at 2x for inspection.



UI MINIATURE

32 x 32 source, shown at 4x for inspection.

Same identity. New composition.
The miniature retains the coral roof, cream wall, mint shutter and welcoming doorway with fewer details.

NATIVE UI SIZE / 32 X 32



A day that breathes.

Temporal color can move smoothly while every spatial edge stays made of pixels.

**SUNRISE**

0-30s / 30 seconds

**DAY**

30-150s / 120 seconds

**SUNSET**

150-180s / 30 seconds

**NIGHT**

180-240s / 60 seconds

One saved clock. One scene wash. The cycle stays 240 seconds. Maximum night tint is 22%; compact warm building lights follow the wash. Sky images here are reduced uniformly for editorial comparison.

The UI belongs to the village.

Cream surfaces, warm outlines and native pixel icons. Keep the world visible and the next action legible.

APPROVED NATIVE ICONS / 32 X 32



Move



Stone pick



Iron pick



Steel pick



Titanium pick



Craft



Details



Wood



Stone



Dirt



Deepslate



Leaves



Obsidian



Bucket



Water



Lava

FUTURE CONTROL STATES / SCHEMATIC, NOT A SHIPPED REDESIGN



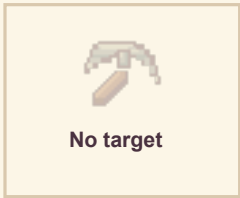
Rest



Hover



Selected



Unavailable

Use more than color.
Selection has an edge or marker. Focus stays visible. Warnings pair a native icon with short text. Tooltips and accessible labels carry names.

Panels open beside their toolbar trigger. On narrow screens, adapt panel placement and hit areas while keeping 32px icons native. Use short labels, useful numbers, clear quantities and a stable focus order.

Readable by design.

STANDARD FOR NEW WORK / These are project targets. Existing controls have not all been migrated or audited.



44 X 44

CSS px touch target

32 x 32 artwork remains native inside a larger interactive area. Schematic: one CSS px shown as one PDF point; actual viewing size varies.

14-16 px

Body and help text

Use a 4/8/12/16px spacing rhythm. Reflow panels before reducing icon size. Keep focus distinct from selection, and explain unavailable actions with a readable reason.

12 px+

Essential quantities

OPAQUE SRGB TOKEN PAIRS / NORMAL TEXT TARGET 4.5:1

Aa 012 9.64:1 Preferred body text	Aa 012 7.12:1 Selected-row text	Aa 012 6.40:1 Warm secondary text
Aa 012 3.77:1 Below text target	Aa 012 3.99:1 Below text target	Aa 012 4.15:1 Below text target

W3C: 4.5:1 normal text; 3:1 large text / meaningful non-text edges. WCAG AA target minimum is 24px with exceptions; 44px is our stronger project standard. Full references and exceptions: ART-BIBLE.md sections 4 and 9.

Text contrast

Control contrast

Target size

Use of color

From rooted meadow to open sky.

EXPLORATION / A visual roadmap for the tower's upward ambition, not a promise of new biomes or mechanics.

Rooted meadow

CURRENT BASELINE / Care, shelter and a workable beginning. Rich loam, gentle hills, coral roofs and warm windows.

Deep workshops

Cool layered rock, timber braces and tiny hearths. Keep deposits and safe open space readable.

Cloud frontier

Quiet sky, stepped cloud masses, sparse silhouettes. Clouds are not implied platforms.

High scaffold

Long block-built verticals against open sky. Decoration must never invent support or wind physics.

Heaven / confrontation

Severe repeated forms, immense negative space, concentrated cream/gold light. Keep Workers handmade.

Change mood through composition and material proportions first. New ramps require a shared-palette review.

Movement should explain the event.

PRESCRIPTIVE STANDARD / New effects communicate cause, material and result without obscuring the world.

01

CONTACT

A precise event

The tool meets the correct surface on the simulation strike.

02

RESPONSE

A material clue

A few coherent clusters inherit the struck material ramp.

03

RECOVERY

A quiet return

Particles disappear promptly; the next action stays readable.

CURRENT ACTIVITY LOOPS

Fire	4 frames x 180 ms	Compact warm fire mouth
Blacksmith hammer	6 frames x 135 ms	Visible working rhythm
Smoke	4 poses x 280 ms	Stepped motion, no soft blur

Deferred: terrain impacts, rewards, ladder art, death and revival, status feedback, weather and new Worker actions need their own review. Decorative loops never alter crafting duration or resource output.

Make every new asset easy to judge.

Read the canonical bible, inspect the native gallery, then write the smallest brief that preserves the rules.

ASSET BRIEF

Purpose

What action or state must be recognized?

Canvas + anchor

Native size, world/UI role and ground line.

Palette + light

Existing named ramps and upper-left light.

Neighbors + states

What touches it? What changes over time?

Constraints

Footprint, collision, saves and simulation.

Review scene

Native view, day/night, UI and motion.

ACCEPTANCE GATE

01 Reads at 1x

Silhouette and purpose survive native display.

02 Fits the family

Shared palette, pixel scale, contour and material density.

03 Behaves honestly

Anchors, support, quantities and strike timing match the game.

04 Works in context

Day/night, ore, motion, open inventories and phone layout.

05 Ships with evidence

Gallery, focused regressions, native screenshots and inventory entry.

Approve the family, then expand.

A new effect or age variant is not approved simply because its palette matches.

Keep the bible alive.

This PDF is the illustrated reading companion. The repository files below carry the complete, maintained rules.

REPOSITORY DIRECTORY / tower-of-babel/buttonwood/

ART-BIBLE.md	Canonical art direction and future visual standards.
ART-GUIDE.md	Implementation details, timings and approved native behavior.
INVENTORY.md	What is implemented, what is deferred and what needs review.
palette.js	Exact named colors and shared Worker animation timings.
index.html	Native gallery; inspect size, silhouettes and motion.

RESEARCH / SELECTED PRIMARY REFERENCES / FULL NOTES IN ART-BIBLE.MD

Re-Logic / Terraria screenshots

Scene hierarchy and purposeful local detail.

Re-Logic / Terraria 1.3 UI upgrades

Historical note: hover, counts and open states.

Mojang / Try the new Minecraft Textures

Jasper Boerstra on crisp, coherent textures.

Mojang / Caves & Cliffs developer Q&A

Whole-space cave composition, including ceilings.

cure / Pixel Joint pixel-art tutorial

Intentional clusters, ramps, banding and noise.

SLYNYRD / Pixelblog 28: Side View Tiles

Tiling and variation; keep our front elevation.

SLYNYRD / Pixelblog 8: Intro to Animation

Key poses, timing and motion review.

SLYNYRD / Pixelblog 26: UX/UI Design Basics

A coordinated visual language for interface assets.

Edition 01 / September 26, 2026 / Canonical revision: 8f1fb3492c89. All pictured game art comes from this repository. Research is translated into original Buttonwood rules; no third-party game assets are embedded.